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# Pattern Arithmetic: Verdicts from Bowls and Paths

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September 18, 2026

## Abstract

A companion paper, *Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere*, defines a unit of truth  $x = (r, \hat{n}, \chi, \tau)$  on a sphere  $\mathcal{S}$  and an algebra over such units — gather ( $\oplus$ ), remove ( $\ominus$ ), mix ( $\boxplus$ ), walk ( $T$ ), and offset — together with an admissibility test that decides which sorts  $\tau$  may use which operators truthfully. That paper builds the algebra and stops at the reading: a bowl of units may be mixed or walked, and the result is a state. It does not say how a state, or a bowl, becomes a *decision*.

This paper adds the missing step. A shaman, a tribal meteorologist, or a market analyst holds a handful of signs and must arrive at one of a small number of conclusions — water or no water, storm or calm, rise or fall. We show that the algebra already contains two distinct mechanisms for this, corresponding to the two readings the first paper leaves undemonstrated: an order-free reading, in which evidence accumulates as phasors and a balanced case returns no answer rather than a forced one; and an order-dependent reading, in which evidence is walked as a sequence of actors to a landing compared against seated names. We give a rule — the *swap test* — for which reading a given body of evidence calls for, and identify four components a decision needs beyond what the algebra supplies: a comparison rule, a selection rule for which signs enter and in what order, a principled abstention, and a place for a lexicon of conclusions to be learned rather than declared. This paper is a framework: it specifies these four components precisely enough to be filled in, and works one example — the diviner’s shells — by hand. It does not report an evaluation against data, and says exactly where that evaluation would have to be made.

# 1 Introduction

A shaman throws ten shells and reads a future from them. He may throw twenty and read it better, or forty better still — but unless he says so, no one else knows what he saw. The shells are evidence; the saying is the verdict. Pattern Arithmetic gives a precise account of the shells. It has none, so far, of the saying.

This is deliberate on the first paper’s part, not an oversight. That paper’s discipline was to make the algebra trustworthy before asking it to decide anything: an admissibility test that refuses sorts with no honest map onto the operators, a type table that says which combinations are legal, and a running insistence that a walk’s landing is compared to a name, never confused with one. Asking that paper to also commit to a decision rule would have asked it to defend a claim it was not ready to make.

It is ready now. This paper takes the atom, the bowl, and the path as given — cited, not re-derived — and asks the next question: given a bowl of evidence, what is the answer?

## 1.1 A human can hold a handful; a machine need not stop there

A diviner’s working memory bounds how many shells, how many birds, how many signs can be weighed against one question at once. Nothing in the algebra of the first paper imposes that bound. The claim of this paper is narrower than it might sound: not that more evidence is always better, but that *the mechanism for combining evidence does not change shape as the amount of evidence grows* — the same gather, the same two readings, apply to ten shells or ten thousand sensor readings. Where that mechanism strains under scale (§4.3) is stated plainly rather than smoothed over.

# 2 What Is Inherited

We use, without re-deriving, the following from the companion paper. Equation and axiom numbers refer to that paper.

- The atom  $x = (r, \hat{n}, \chi, \tau) \in \mathcal{S}$ , with  $r \leq 1$  a normalized strength,  $\hat{n} \in S^2$  a direction,  $\chi \in [0, 2\pi)$  a phase, and  $\tau$  a sort tag that gates which operators may be truthfully applied (Axiom 1; Axiom 0).
- Gather,  $\oplus$ : union on finite subsets of  $\mathcal{S}$ , sort-blind, idempotent, lossless, undone by  $\ominus$  (Axiom 4).
- Mix,  $\boxplus$ : a phasor reading of a bowl whose members share a direction and a sort,  $\boxplus(B) = (\min(1, |z_B|), \hat{n}, \arg z_B, \tau)$  with  $z_B = \sum_{x \in B} r_x e^{i\chi_x}$ ; commutative, non-associative, lossy, uninvertible (Axiom 5).
- The walk,  $T$ : an ordered composition  $T_{w_n} \circ \cdots \circ T_{w_1}(x)$  of actor-parameterized rotations and phase shifts; non-commutative in general, and many-to-one, so a landing alone does not determine the walk that reached it (Axiom 6; §3).
- *Levels are closed*: a unit carries no representation of what it constitutes, and a bowl carries no representation of the reading that will be applied to it. Whatever a lower unit would need to know about a higher one — which units were taken as

co-present, over what interval, under what question — is a constituting fact of the higher level, not a coordinate of the atom (companion paper, §2.2).

Two consequences of level closure matter enough here to restate. First, a bowl that saturates under Mix is often a level that has been flattened — repeated samples of one unit, mistaken for many units at one level (companion paper, §2.7). Second, the atom stays a four-tuple; grouping criteria such as a time window or a spatial radius live outside it, deciding bowl membership before the algebra runs rather than riding inside any single unit.

### 3 Two Readings of One Bowl

Given a bowl  $B = \{x_1, \dots, x_n\}$  of evidence bearing on one question, the companion paper currently defines two readings that turn  $B$  into a single state: phasor Mix and ordered transformation. This is not a claim that no other reading could be defined; it is the pair that paper supplies today, and the pair this paper uses. We name them here because the first paper defines both and demonstrates only one.

#### 3.1 Reading B: the walk

Seat the question as a state  $q$ . Apply the members of  $B$  as actors, in some order  $\pi$ :

$$\text{land}_\pi(q) = T_{x_{\pi(n)}} \circ \dots \circ T_{x_{\pi(1)}}(q).$$

The verdict is the landing, compared against seated names (the storm’s calm and storm pins are an instance of this). Order matters by construction:  $T_a T_b \neq T_b T_a$  in general, so  $\text{land}_\pi \neq \text{land}_{\pi'}$  for  $\pi \neq \pi'$  is the expected case, not a failure. This reading is fully worked in the companion paper’s storm example, which we do not repeat.

#### 3.2 Reading A: the phasor sum

Rotate every member of  $B$  onto the question’s axis  $\hat{n}_q$  (a rotation, which preserves  $r$  and  $\chi$ ), then read the whole bowl at once:

$$\boxplus(B) = \left( \min(1, |z_B|), \hat{n}_q, \arg z_B, \tau \right), \quad z_B = \sum_{x \in B} r_x e^{i\chi_x}.$$

Order plays no role:  $z_B$  is a sum, and sums do not care about the order of their terms. The verdict’s *confidence* is  $|z_B|$ ’s image under the cap, and its *direction* is  $\arg z_B$ , read against a lexicon of conclusions seated at particular phases (e.g.  $\chi = 0$  for “water,”  $\chi = \pi$  for “no water”). Agreeing evidence lengthens the sum; opposing evidence shortens it; evenly opposed evidence returns  $z_B = 0$  — black, and a principled abstention rather than a forced choice (companion paper, Axiom 2).

#### 3.3 The swap test

Which reading applies to a given  $B$  is decided by one question: does exchanging the order in which two members of  $B$  arrived change the answer?

- If **no** — the shells thrown together, ten readings taken at one instant —  $B$  has no order to respect, and Reading A is the correct mechanism. Applying Reading B here manufactures an order-dependence that is not in the evidence.
- If **yes** — birds fleeing, then a wind shift, then a temperature drop, where the same three signs in a different sequence mean something else —  $B$ 's order is itself information, and Reading A would discard it. Reading B is required.

The two readings are not competing hypotheses about one mechanism; they are the correct tool for two different shapes of evidence, and the swap test is how a system — human or otherwise — tells which shape it is holding.

### 3.4 The readings compose

Nothing prevents using both on one episode. A morning's observations may arrive in a meaningful sequence (Reading B over the morning) and also include several simultaneous signs at one moment within it (Reading A over that moment's bowl, itself one actor in the larger walk). This is level closure in practice (§2): a Reading A verdict is a state, and a state may act as  $w$  in a  $T$ -path, exactly as any other seated unit may.

## 4 Four Components the Algebra Does Not Supply

Both readings above take a bowl and return a state. Neither, on its own, returns a decision a person or a system can act on. Four components stand between a state and a verdict, and none of them is supplied by  $\oplus$ ,  $\boxplus$ ,  $T$ , or offset.

### 4.1 A comparison rule

The companion paper's storm section says a ball falls *near* the calm pin or the storm pin, and does not define near. A working comparison rule must state, at minimum: which coordinates of the landing are read, what metric is used on those coordinates, and whether  $\tau$  must match exactly or may differ.

Two states are compared only if  $\tau_A = \tau_B$ ;  $\tau$  gates comparison the same way it gates every other operator. Given that, a distance is taken over all three geometric coordinates, each measured the way the companion paper already measures it elsewhere rather than by naive coordinate subtraction:

$$d(A, B)^2 = w_r (r_A - r_B)^2 + w_{\hat{n}} \arccos(\hat{n}_A \cdot \hat{n}_B)^2 + w_{\chi} \left( \min(|\chi_A - \chi_B|, 2\pi - |\chi_A - \chi_B|) \right)^2, \quad (1)$$

using the geodesic distance the offset operator already relies on for  $\hat{n}$ , and the circular distance a phase requires for  $\chi$ ;  $r$  alone is a bounded line with no wraparound and needs neither correction. Raw coordinate subtraction on  $\hat{n}$  or  $\chi$  is *not* a safe default: two directions or phases near 0 and near  $2\pi$  are adjacent on the sphere or the circle but appear maximally distant under naive subtraction, and reference atoms placed near a natural zero point — which is common, not rare — would be misjudged systematically rather than occasionally. *Open in this paper:* the weights  $w_r, w_{\hat{n}}, w_{\chi}$ , meaning how much a mismatch in strength, direction, or phase should count against an otherwise close match. §5 does not need this choice made in general: every shell and every reference point there already shares  $\hat{n}_q$  by construction, so  $w_{\hat{n}}$  never enters, and the verdict is read

from  $\chi$  (which reference point is nearest) with  $r$  supplying the null rather than a weighted distance. A question with reference atoms at different directions would need  $w_{\hat{n}}$  and  $w_{\chi}$  actually chosen against each other, which remains open.

## 4.2 A selection rule

For Reading B, the order  $\pi$  that the evidence is applied in is handed to the walk; it is not the walk’s to choose. A real system must decide which signs enter  $B$  at all (§4.1’s window and radius criteria, companion paper §2.2) and, when order matters, in what sequence. This selection is itself where most of an inference system’s actual difficulty lives, and it is entirely outside the algebra. *Open in this paper.*

## 4.3 A principled null

Reading A supplies one for free:  $z_B = 0$ . Reading B does not. A walk always lands somewhere, and a comparison rule (§4.1) always returns a nearest pin, however far away it is. A Reading-B system therefore needs an explicit distance threshold beyond which the verdict is *abstain* rather than *nearest pin regardless*. This is a second place, distinct from §4.1, where a number must be chosen rather than derived.

## 4.4 Learned pins and learned signs

The companion paper notes that a lookup  $E_L$  may be learned (§5.6) without the units it points into being anything other than fixed. The same discipline applies to inference: what is learned is narrow. A sign’s  $(r, \chi)$  for a given question — how strongly, and in which direction, a particular bird’s flight bears on “storm” — is an  $E_L$  entry, learnable from labelled cases. The seated pins storm, calm are likewise  $E_L$  entries, and may be learned rather than declared. Neither requires learning a representation in the sense of an embedding; both are learning one small map, per sign, per question. *This is the paper’s one concrete claim about the learning loop:* it is cheaper than representation learning because it is narrow, not because it is avoided.

**Building a reference atom.** A pin need not be declared; it can be built from witnessed cases the same way live evidence is read. For a class  $c$  (one storm, one drought), let  $G_c = \{y_1, \dots, y_m\}$  be the Reading-A composite of each witnessed episode, seated by  $E_L$  from that episode’s own bowl and already rotated onto the question’s axis  $\hat{n}_q$  (§3.2). Because every  $y_i$  shares  $\hat{n}_q$  by construction,  $\boxplus(G_c)$  is defined by the companion paper’s bowl-Mix equation (§2.7) without needing its still-open case of Mix across differing directions (its Future Work); building a reference atom this way never depends on a question that paper leaves unresolved. The reference atom for  $c$  is  $\text{ref}_c = \boxplus(G_c)$ .

**Coherence, and when to split.** The strength  $r$  of  $\text{ref}_c$  is not only a confidence value; it is a coherence check on  $c$  itself. Members of  $G_c$  that agree add constructively; members that genuinely disagree — because  $c$  is not one pattern but two, wearing one label — partially or fully cancel, exactly as opposed evidence cancels in §5. A weak  $\text{ref}_c$  is therefore not noise to smooth over; it is the mechanism reporting that  $c$  should be split. Given a coherence threshold  $\theta \in (0, 1)$  and a minimum group size  $k_{\min}$  below which a split is not attempted: if  $r(\text{ref}_c) \geq \theta$ , keep  $\text{ref}_c$ ; otherwise, if  $|G_c| \geq k_{\min}$ , partition  $G_c$  into

sub-groups and recurse on each, so that a single label may resolve into several reference atoms (a slow-onset and a flash drought, say); otherwise accept  $\text{ref}_c$  as weak, or fall back to comparing incoming evidence against  $G_c$ 's individual members rather than their compression. How  $G_c$  is partitioned when a split is warranted is not specified here and is left open, as is the choice of  $\theta$  itself;  $\theta$  must be bounded away from 0, since every singleton group is trivially coherent and a threshold with no floor never stops splitting. Group size strictly decreases at each split, so the recursion terminates regardless of  $\theta$ 's exact value.

**This is level closure, applied to references.** A class discovered to need splitting is a level-2 object in its own right (companion paper, §2.2): the split is discovered from  $G_c$ 's own internal agreement, not asserted in advance, and each resulting sub-reference is exactly as usable — comparable, walkable, further splittable — as the class it came from. A domain with genuinely few witnesses (two droughts, not nine) is not a failure of the procedure;  $k_{\min}$  says plainly that there is not yet evidence to subdivide responsibly, and the honest output is a weaker reference or a fall back to individual comparison, not a manufactured split.

**Illustration.** Nine witnessed episodes at  $r = 0.6$  each, at  $\chi = 8^\circ, 9^\circ, 9^\circ, 10^\circ, 10^\circ, 10^\circ, 11^\circ, 11^\circ, 12^\circ$ , compress to  $\text{ref} = (1.00, \hat{n}_q, 10.0^\circ)$ : coherent, confidently kept. The same nine witnesses, five at  $\chi = 0^\circ$  and four at  $\chi = 180^\circ$  — two real sub-types wearing one label — compress to  $r = 0.60$ ; a perfectly even split, five and five, compresses to  $r = 0.00$ , the same cancellation that gives §5 its abstain, now flagging the class itself rather than a single reading of it.

## 5 Worked Example: The Diviner's Shells

We work Reading A in full, since the companion paper never carries a phasor reading past two states, and its property demonstration (non-associativity, on an anonymous triple) is not a reading of a bowl that means something.

Seat the question *will it rain* at  $\hat{n}_q$ . Each shell, once cast, is read by a diviner (or, later, by a learned  $E_L$ ) as a state  $(r, \chi)$  at  $\hat{n}_q$ :  $\chi = 0$  favors rain,  $\chi = \pi$  favors no rain, and intermediate  $\chi$  is a hedge. Five casts, each shell carrying  $r = 0.4$  unless noted:

| Cast                                           | reading                                 | interpretation         |
|------------------------------------------------|-----------------------------------------|------------------------|
| 4 shells, all favor rain                       | $r = 1.00$ at $\chi = 0^\circ$          | confident rain         |
| 3 favor, 1 against                             | $r = 0.80$ at $\chi = 0^\circ$          | rain, less confident   |
| 2 favor, 2 against                             | $r = 0.00$                              | <b>abstain</b>         |
| 4 favor, 2 hedge at $90^\circ$                 | $r = 1.00$ at $\chi \approx 26.6^\circ$ | rain, hedged direction |
| 12 shells ( $r = 0.15$ each), all weakly favor | $r = 1.00$ at $\chi = 0^\circ$          | confident rain         |

The third row is the case worth dwelling on. Two shells favor rain and two oppose it, at equal strength; the phasors cancel exactly, and the verdict is black — not a coin flip, not a default to the majority class, but the algebra's own statement that this evidence does not decide the question. A system built on a forced classifier (softmax over two classes, say) has no way to express this outcome; it must output a confident-looking number even when the evidence is silent. Reading A's null is not an afterthought fitted

onto the algebra; it is Axiom 2, doing here exactly what it does for a Mix reading of colour.

The fifth row shows the saturation the companion paper’s §2.7 discusses: twelve weak, agreeing signs pin at  $r = 1$  exactly as four strong ones do, and the difference between “moderately confident” and “overwhelmingly confident” is lost. This is the place where §4.2’s selection rule matters most: whether those twelve readings are twelve units at one level, or repeated samples of one unit that should have been reduced (companion paper, Axiom 4) before ever reaching the bowl, is a question this paper does not resolve and flags as the clearest candidate for the next.

*Reading B on the same shells* is not worked here. Doing so honestly requires an order for the casts, which a physical handful of shells thrown at once does not supply — exactly the case the swap test (§3.3) assigns to Reading A. A worked Reading B example belongs to evidence that genuinely arrives in sequence, and the companion paper’s storm already provides one.

## 6 Level Closure in Practice: Correcting a Path

A  $T$ -path records the sequence of actors applied to it, faithfully, by definition (companion paper, Axiom 6). It has no means of knowing whether that sequence was the right one — level closure (§2) says precisely this: a lower unit carries no representation of what it constitutes, and neither the atom nor the path can see the higher-level context that would tell it it has erred.

A concrete case: a sequence of observed signs is walked in the order observed, but a later sign (or a completed higher-level unit, such as the rest of a sentence or the rest of a song) makes that order appear wrong — *twinkle, little, twinkle, star* rather than *twinkle, twinkle, little, star*. Correcting this is a level-2 operation: it requires comparing the recorded path against context the path itself does not have access to, and revising the walk probabilistically. This paper’s position is that the path must therefore be *legible* to the level above — readable as an ordered record, not only walkable — so that a corrector at the next level can inspect and revise it. Nothing in the companion paper’s axioms prevents this; nothing in them provides it either. We flag it as the clearest point of contact between this paper’s verdicts and a learning loop operating above them, and leave the corrector itself unspecified.

## 7 Scope and Open Questions

This paper is a framework, in the same sense the companion paper is: it specifies mechanism and states honestly what it has not demonstrated at scale. [*OPEN — decide before circulating: does this paper stop here, as a specification plus one worked toy example, or does it commit to an evaluation against real data (e.g. a labelled forecasting or forex task) with a stated baseline? The two choices imply different remaining sections and different claims in the abstract; see the companion paper’s own title history for how much that choice shapes what a title may honestly claim.*]

Regardless of that choice, the following are stated open rather than resolved by this paper, and are drawn directly from §4:

- The comparison rule (§4.1) is a choice, not a derivation; a different metric may be more honest for a given domain.

- The selection rule (§4.2), including how the grouping window of companion-paper §2.2 is set, is the largest source of real difficulty and is entirely unaddressed here.
- The saturation-under-aggregation problem (§5, companion paper §2.7 and §2.3) is diagnosed, not solved: whether to route repeated samples through a reduction before the bowl, or revisit the companion paper’s Axiom 1 note on an order-preserving bijective bound, is left to whichever setting first needs to aggregate at scale.
- The path corrector (§6) is named and not built.